Four cell-filter models

20 donors · **37 markers** · `gated_alive`

100 shared outer splits: 14 training / 6 test donors.

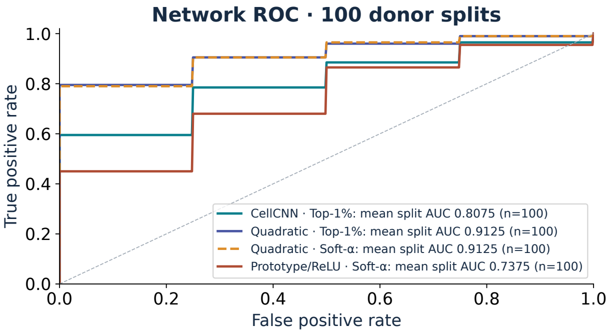
Model	Cell response $h_k(x)$	Pooling
CellCNN	$\mathrm{ReLU}(b_k + w_k^\top x)$	hard Top-1 %
Quadratic Top-1 %	$\mathrm{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	hard Top-1 %
Quadratic Soft-α	$\mathrm{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	learnable Soft-α
Prototype Soft-α	$\mathrm{ReLU}(\rho_k - \frac{1}{d} \sum_j a_{kj} (x_j - c_{kj})^2)$	learnable Soft-α

Population: largest positive output contrast; cells with $h > \frac{1}{2} \max_{\text{inner-train}} h$.

Donor prediction and CMV-status association of the selected population are evaluated separately.

Cohort: 9 CMV+, 11 CMV−. Each split: train 7+/7−, test 2+/4−. ArcSinh(x/5); scaling and half-max reference use the selected model's inner-training donors only.

Network classification across 100 donor splits



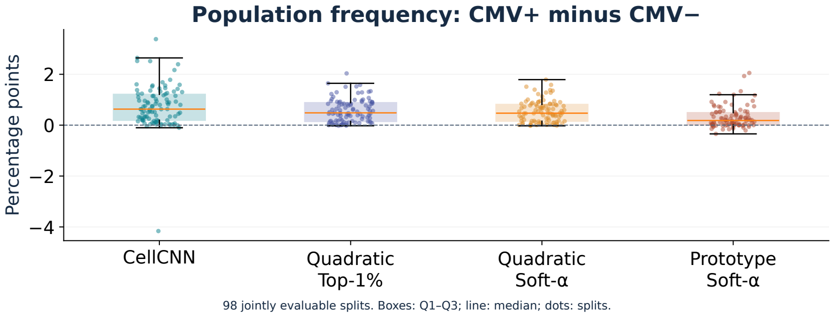
AUC

Model	Mean	Median
CellCNN	0.8075	0.8750
Quad. Top-1%	0.9125	1.0000
Quad. Soft-α	0.9125	1.0000
Prototype	0.7375	0.7500

Quadratic Top-1% and Soft-α share the highest mean network AUC (0.9125).

100 outer test sets, each with 2 CMV+ and 4 CMV− donors. Donor score: mean of five prediction bags of 20,000 cells. Curves are averaged per split; legend values are mean original split AUCs.

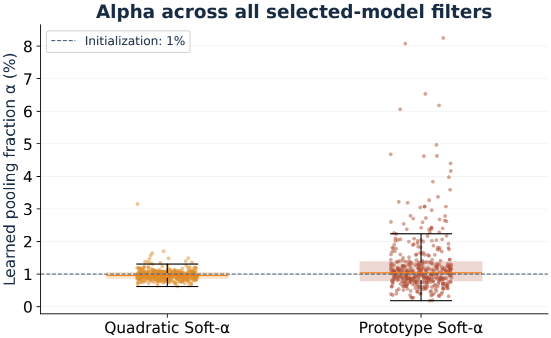
Selected-population frequency difference



Largest mean frequency difference: CellCNN (0.740 percentage points).

98 jointly evaluable splits; test set: 2 CMV+ and 4 CMV− donors. One fixed 20,000-cell sample per test donor. Effect = mean CMV+ frequency minus mean CMV− frequency, in percentage points.

What does the extra pooling parameter learn?



Rank fraction, not population frequency

Alpha weights cell ranks within a bag. Half-max defines the separately evaluated population.

Median α: Quadratic 0.96%; Prototype 1.04%. Alpha controls pooling, not measured population frequency.

All filters from 100 selected models per variant: 416 Quadratic and 414 Prototype filters. These are model parameters, not independent donor observations.

Take-away

- Quadratic Hard and Soft share the highest mean network AUC on this dataset.
- Soft vs. Hard: 5 better, 90 tied, 5 worse; mean ΔAUC 0.0000.
- Quadratic Soft-α vs. CellCNN: 53 better, 37 tied, 10 worse; mean ΔAUC $+0.1050$.
- Only 20 independent donors: repeated splits are descriptive, not independent replication.

Quadratic Hard and Soft both reach 0.9125 mean network AUC; learnable soft pooling does not improve it here.

All 100 planned outer splits are included. Paired differences use the same test donors in each split. These results do not establish performance on a new cohort.

Backup: half-max population frequency

Filter selection and fixed training threshold

$$\Delta_k = V_{1k} - V_{0k}, \qquad k^* = \arg \max_{k:\Delta_k>0} \Delta_k$$

$$M = \max_{x \in \text{inner-train cells}} h_{k^*}(x), \qquad t = \tfrac{1}{2} M$$

$$f_i = \frac{\#\{x \in \text{test sample } i : h_{k^*}(x) > t\}}{N_i}$$

One fixed 20,000-cell sample per test donor. The strict rule also applies when $M = 0$.

Two population metrics

$$\mathrm{AUC}_{\mathrm{freq}} = \text{AUC}(y_i, f_i)$$

$$\mathrm{Effect} = \bar{f}_{\mathrm{CMV+}} - \bar{f}_{\mathrm{CMV-}}$$

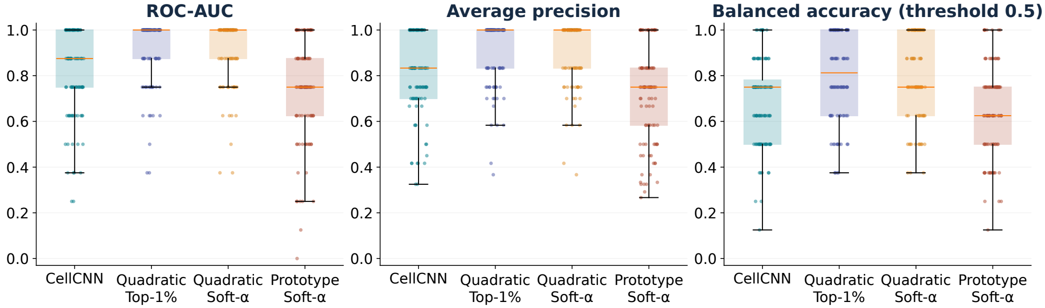
Without a positive filter, values remain missing. No post-hoc flipping of AUC direction.

Frequency comparisons use **98** jointly evaluable splits.

Half-max is cell selection, not clustering; CMV is a donor label, not a cell-type label.

The baseline has no positive-output filter in splits 44 and 84. These two splits are excluded from all common frequency comparisons, but retained for network metrics.

Backup: variation in classification across splits

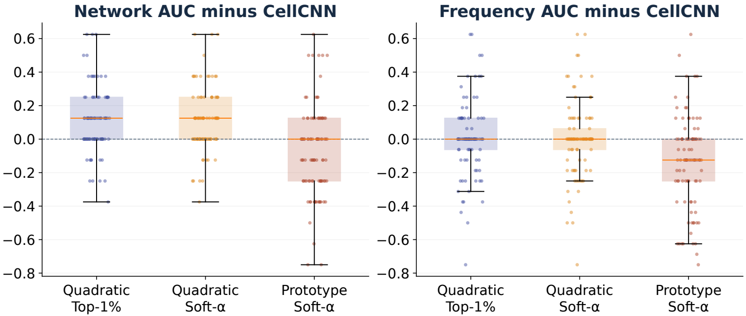


100 overlapping donor splits. Boxes: Q1–Q3; line: median; dots: splits. Not confidence intervals.

Highest means: Quadratic Top-1% for AP (0.907) and BA (0.801).

100 test sets of six donors (2 CMV+, 4 CMV−). AP: average precision; BA: balanced accuracy at score threshold 0.5. Boxes: Q1–Q3; line: median; dots: splits. Not confidence intervals.

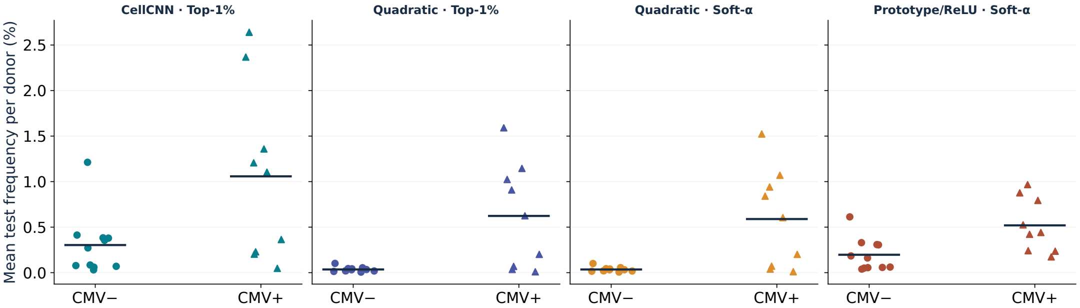
Backup: paired differences to CellCNN



Quadratic Soft-α vs. CellCNN: 53 better, 37 tied, 10 worse; mean ΔAUC $+0.1050$.

Network: 100 paired splits; frequency: 98 common paired splits. Difference = model AUC minus CellCNN AUC within the same split. Positive values indicate higher AUC, not independent replication.

Backup: population frequencies of individual donors

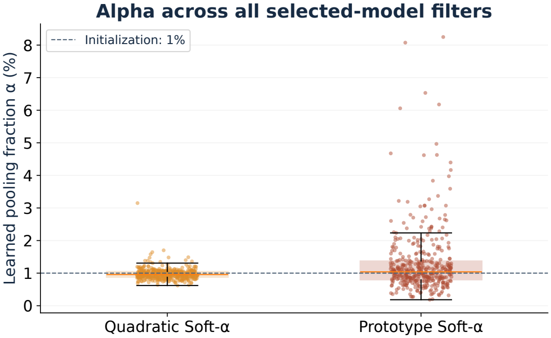


One point per donor: average across available test visits in the common splits. The selected population may vary by split.

Each donor contributes one point, averaging their available test visits first.

20 donors per model: 9 CMV+ (triangles), 11 CMV− (circles); visits restricted to the same 98 common splits. Bars: mean of donor points. The selected filter may differ between splits.

Backup: the learnable rank mask



Rank fraction, not population frequency

$$\alpha_k = \sigma(\theta_k), \qquad \alpha_k^{(0)} = 0.01$$

$$m_{rk} = \sigma\left(\frac{\alpha_k - q_r}{0.002}\right)$$

$q_r = (r - \frac{1}{2})/N$ for descending response ranks. Temperature is fixed. Soft $\alpha = 1\%$ is not identical to hard Top-1%.

Median α: Quadratic 0.96%; Prototype 1.04%. Alpha controls pooling, not measured population frequency.

100 selected models per variant; all 416 Quadratic and 414 Prototype filter parameters are shown. The temperature is not fitted or tuned in this comparison.

Backup: scope of the controlled comparison

- **Quadratic Hard vs. Soft:** same cell response and L2 penalty; the pooling function changes.
- **CellCNN vs. Quadratic:** adds a quadratic term and its L2 penalty.
- **Prototype vs. Quadratic:** different geometry, initialization and regularization; not an isolated alpha comparison.
- **No refit:** the selected inner model uses 9–10 training donors, not all 14 outer-training donors.
- **Limitation:** variants were developed after earlier results on these same 20 donors.

L2 coefficient 10^{-4}: CellCNN on W, V; Quadratic on W, Q, V; Prototype on V, plus $10^{-3}\,\mathrm{mean}((a-1)^2)$. No extra alpha or radius penalty.

Quadratic Hard and Soft both reach 0.9125 mean network AUC; learnable soft pooling does not improve it here.

Nine candidates per outer split: three inner folds and 3/4/5 filters. The same donor splits, sampling seeds, learning rate and early-stopping procedure are used.